

VL 201: Semiconductor Devices for Integrated Circuits

Course Code: VL 201

Course Name: Semiconductor Devices for Integrated Circuits

L-T-P-C: 3-0-0-3

Prerequisites: Applied Electronics (IC 161)

Intended for: Discipline Core for B.Tech. in Microelectronics and VLSI Engineering students

Mutual Exclusion: EE 311

Approval: 61st BoA

Preamble

The objective of the course is to provide the fundamental knowledge for understanding the concepts of semiconductor devices for Integrated Circuits (IC's). This course builds the knowledge base of the physics of semiconductors as related to the characteristics and design of solid-state electronic devices for IC's. This is a required core-course for VLSI and Microelectronics Engineering undergraduate students.

Course Modules with quantitative lecture hours

- **Unit-1: Introduction to Semiconductors:** Introduction & Evolution of Semiconductor Devices and Technology; Semiconductor Materials; Crystal Planes & Miller Indices; Band Model of Solids; Concept of Equilibrium, Electrons & Holes; Free Carriers in Semiconductors; Carrier Distribution Functions (Intrinsic & Extrinsic semiconductors) (9 hours)
- **Unit-2: Current Conduction Mechanism in Semiconductors:** Carrier Drift, Measurement of Semiconductor properties (Four-Point Probe & Hall Effect Methods); Carrier Diffusion; Excess Carriers (Generation & Recombination) and Characteristics of Excess Carriers; Thermionic Emission; Tunneling, High Field Effect (8 hours)
- **Unit-3: P-N Junctions:** Fundamentals of p-n junction, p-n junction under thermal equilibrium and electric field, I-V characteristics of a p-n junction, Junction Breakdown: Zener & Avalanche, Small Signal Equivalent Circuit, Devices based on p-n junctions: LED, LASER, Photodetectors & Solar Cells (9 hours)
- **Unit-4: Metal-Semiconductor Contacts:** Ideal M-S Junction, Schottky and Ohmic Contact, Effect of Surface States, Devices based on M-S Contacts: MESFET, MODFET (3 hours)
- **Unit-5: Bipolar Transistors:** Physics and operation of bipolar junction transistors, Current conduction mechanism in bipolar junction transistor, Effects of Collector Bias Variation (Early Effect), Small-Signal Transistor Model, High-Frequency Transistor Model, Devices based on bipolar junction transistor: Thyristor (4 hours)
- **Unit-6: Metal Oxide Semiconductor Field Effects Transistors:** MOS System: The Ideal MOS Structure, Capacitance-Voltage Characteristics, Effect of oxide and interface charges; MOSFET: Structure and operation, Scaling, Short-Channel Effects (V_{th} roll-off, DIBL, Bulk Punch Through); Devices based on MOSFET: CMOS Inverter & Latch up, BiCMOS, MOSFET on Insulator (TFT, SOI), MOS Memory Structures (DRAM, SRAM, NVM) (9 hours)

Text Books

1. S.M. Sze and M.K. Lee, *Semiconductor Devices: Physics and Technology*, John Wiley & Sons, 2012.



References

1. S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, John Wiley & Sons, 2002.
2. D. A. Neamen, *Semiconductor Physics and Devices*, McGraw Hills, 2003.
3. B. G. Streetman and S. Banerjee, *Solid State Electronic Devices*, Prentice Hall, 2005.
4. Robert F. Pierret, *Semiconductor Device Fundamentals*, Addison-Wesley Publishing, 1996.